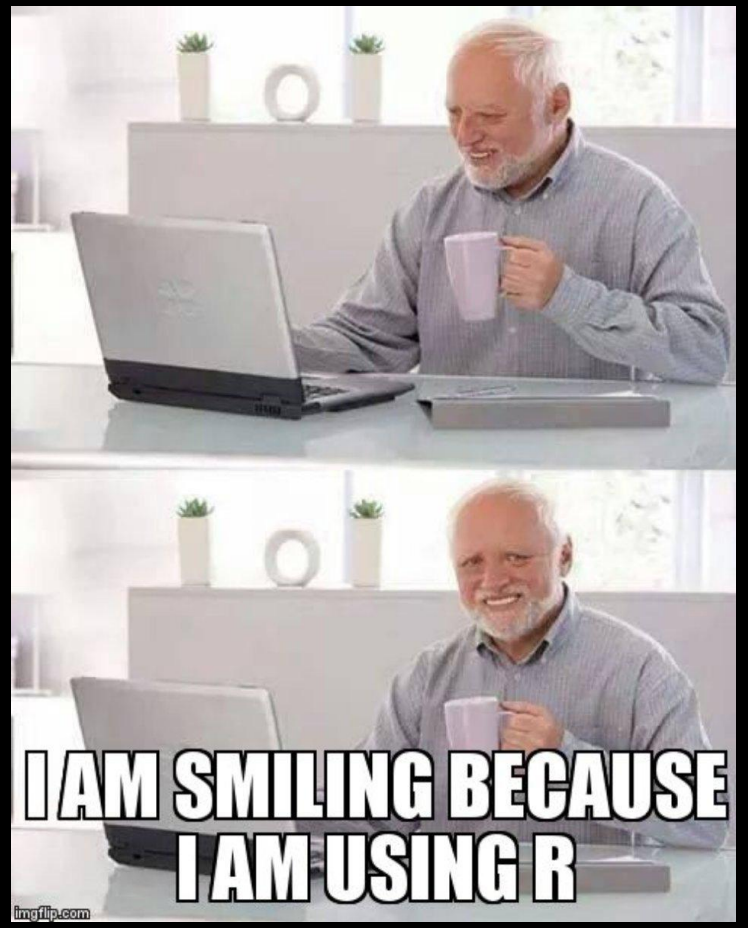


MES 503 Lab 8

Kayla Blincow, PhD

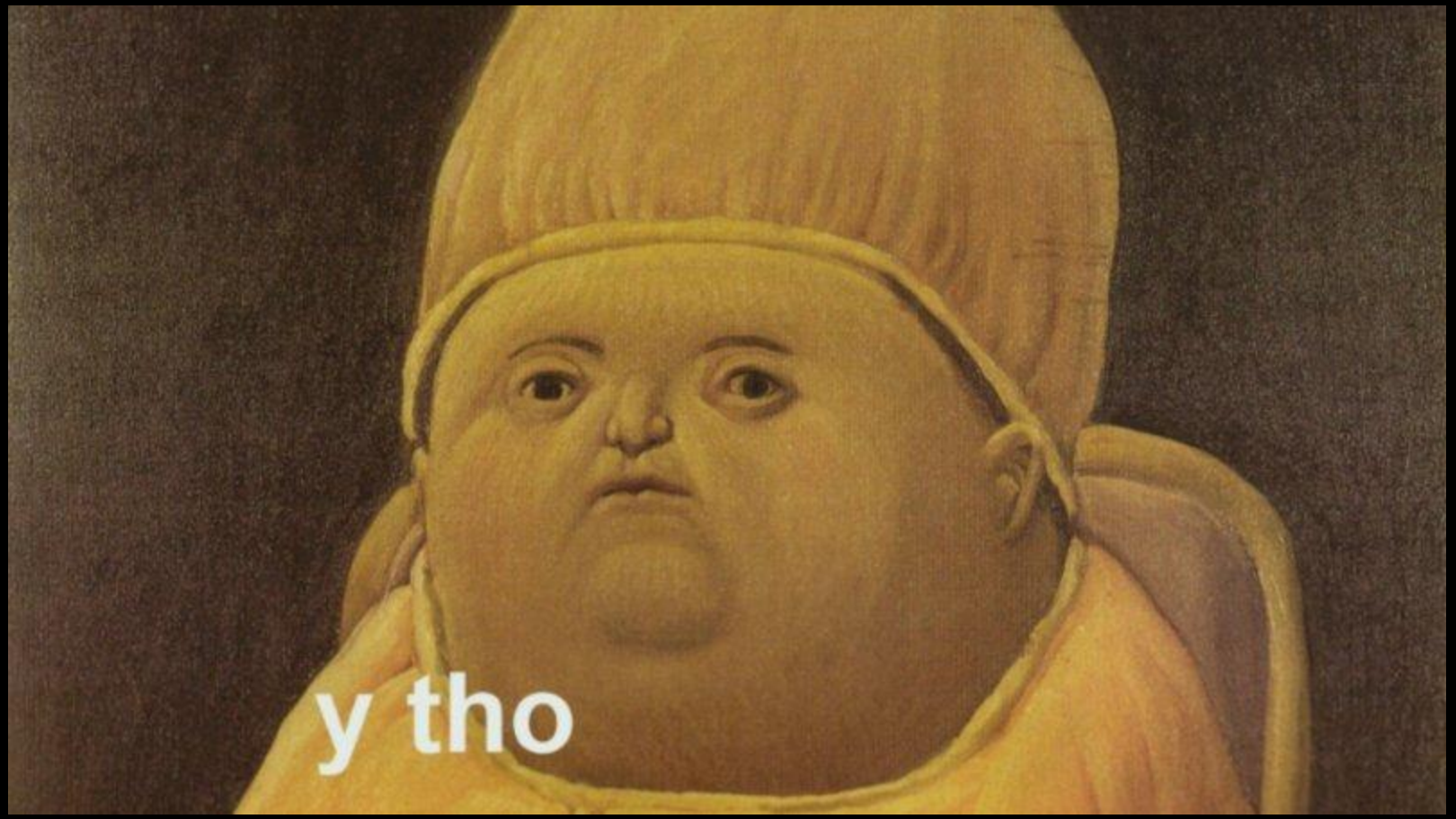
Announcements

- Lab Assignment 2 graded
- New lab assignment posted

TENTATIVE SCHEDULE OF TOPICS – MES 503 Fall 2026

		Mondays (Lectures)		Wednesdays (Labs)	
Week	**Date**	**Lecture Topic**	**Date**	**Lab topic**	**Assessments**
1	8/17	Introduction to Statistics & Experimental Design I	8/19	Introduction to R and RStudio	
2	8/24	Introduction to Statistics & Experimental Design II	8/26	Intro to ggplot2	
3	8/31	Correlation	9/2	Correlation in R	Quiz 1
4	9/7	LABOR DAY (No Class)	9/9	Intro to Data wrangling	Lab 1 Due (9/10)
5	9/14	Statistical Hypothesis Testing & t tests	9/16	t Tests in R	
6	9/21	Linear Regression	9/23	Linear Regression in R	Quiz 2 (9/21) Lab 2 Due (9/24)
7	9/28	ANOVA	9/30	ANOVA in R	
8	10/5	ANOVA II (Two-Way & Nested) (virtual)	10/7	Two-Way & Nested ANOVA in R	Quiz 3
9	10/12	Multiple Linear Regression	10/14	Multiple Linear Regression in R	Lab 3 Due (10/15)
10	10/19	ANCOVA	10/21	ANCOVA in R	Quiz 4
11	10/26	Analysis of Frequencies	10/28	Analysis of Frequencies in R	Lab 4 Due (10/29)
12	11/2	Liberty Day – UVI Closed	11/4	Miscellaneous Fun R Topic	Quiz 5
13	11/9	Quantifying Variability/Figure and Data Sheet Design	11/11	Veterans Day – UVI Closed	
14	11/16	Reading for Stats Presentations	11/18	Reading for Stats Presentations	Lab 5 Due (11/17)
15	11/23	Applying this course to your thesis	11/25	Study Day	
16			12/3	**Final Exam**	

Transforming Data: Why do we do it?



Transforming Data: Why do we do it?

To avoid violating assumptions…

- To even out the variance if the assumption of homoscedasticity is violated
- To normalize a variable if the assumption of normality is violated
- To linearize model if variable looks non-linear
- Reduce the impact of outliers

Let’s Talk Transformations

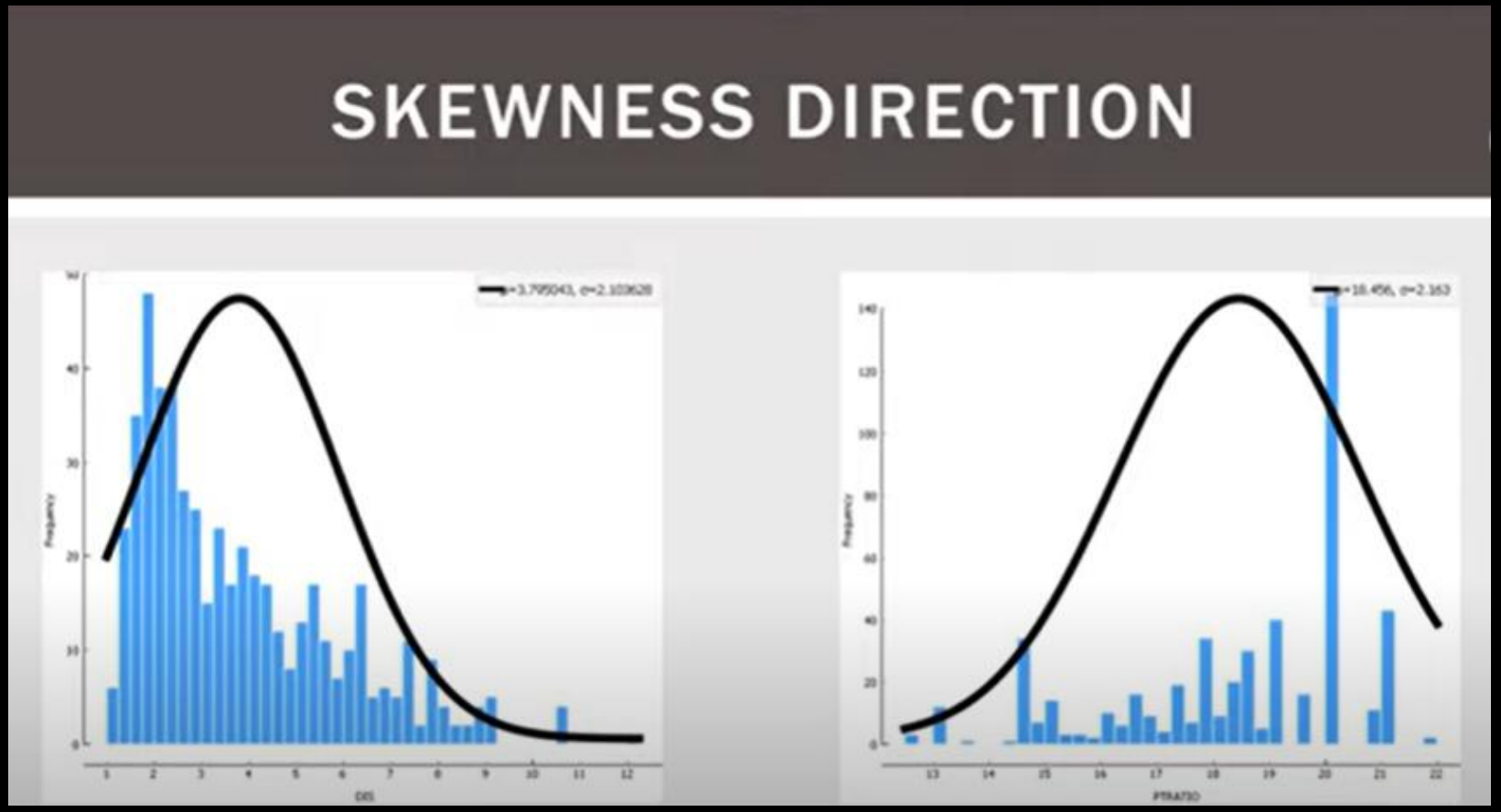
- Transformation changes the scale of the data but doesn’t change the relative ordering of observations
  - E.g. log10() turns values such as 1,10, and 100 into 0, 1, 2 -> the observations are still in the same order, but large values have less influence
- This is okay because typically we are interested in the underlying pattern, rather than the original measurement scale itself

Let’s Talk Transformations

This is NOT changing the data to get the result you want!

The goal of a transformation is not to make the data look better—it is to make the statistical model better suited to the data.

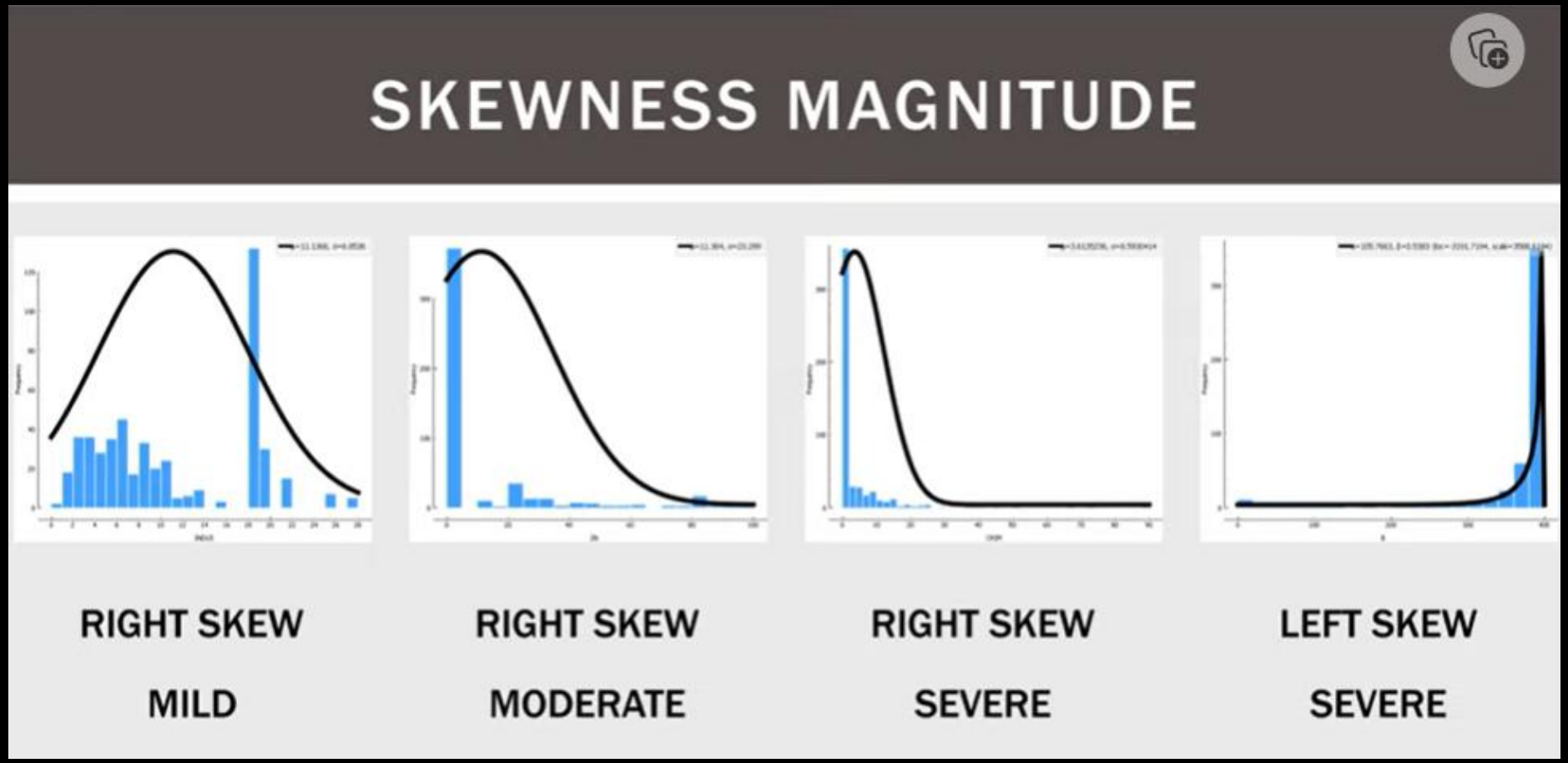
Transforming Data



RIGHT

LEFT

Transforming Data



Transforming Data: Which one to use?

Skew	Moderate	High	(Higher)	Extreme
Positive (right tail)	Square root transformation	Natural log transformation	Log base 10 transformation	Inverse transformation
Negative (left tail)	Reflect then square root transformation	Reflect then natural log transformation	Reflect then log base 10 transformation	Reflect then inverse transformation

- Reflect: flips the distribution, necessary for left skewed data
  - (MAX + 1) – X
- FYI: log() in R is natural log →ln() is the same
  - log10() gives log base 10

Let’s Talk Transformations

When presenting results of a test performed on transformed data:

- State that the data were transformed
- Report the results of the statistical test normally, but make sure you are very clear what your response variable is (log transformed?)
  - Leave no doubt about whether you are talking about raw or transformed data
- Be careful when interpreting effect size
  - You might need to back transform.
- Present figures in a way that helps the reader understand the system (not necessarily the statistics)
  - You may want to show the raw data in figures because the original units are often more intuitive. However, if the statistical analysis was performed on transformed data, make that clear in the figure caption. Alternatively, you can plot the transformed values if that better represents the analysis, but your axis labels should make that clear.
- Interpret your results in the context of the original research question

Analyze on the scale that meets the assumptions but interpret and communicate the result in a biologically meaningful way.

Texas Bluebonnets

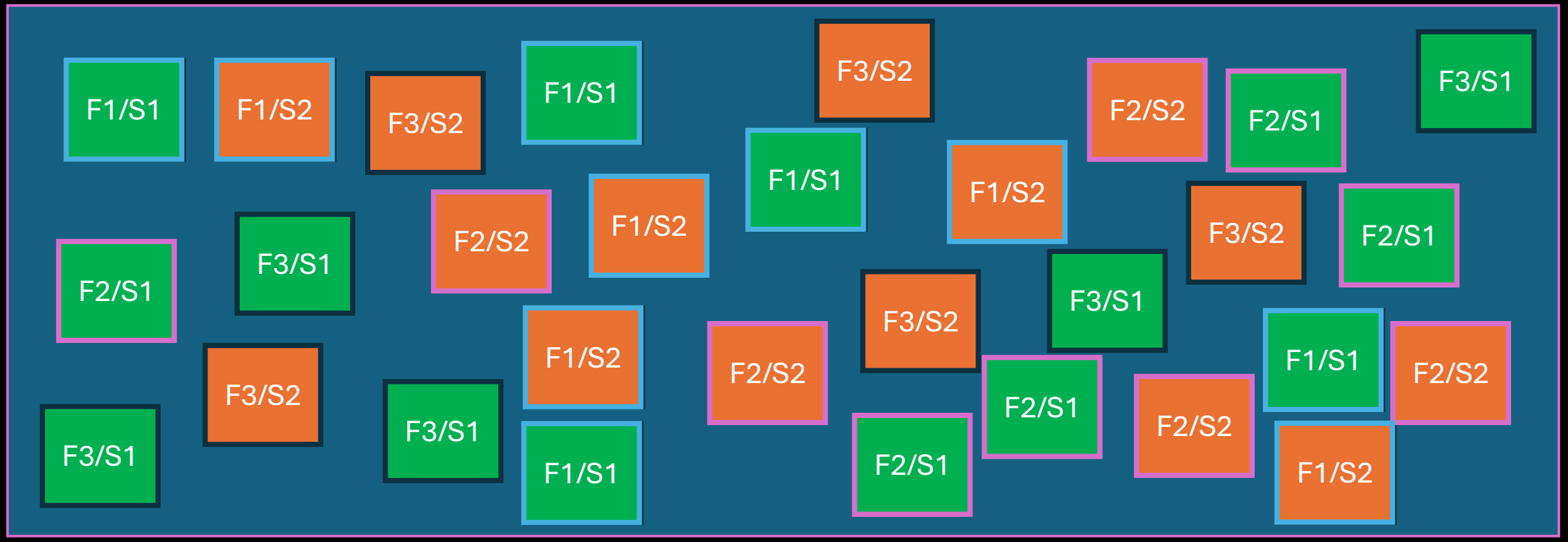


- State flower of Texas
- Let’s say we own a farm with bluebonnets, and Texan Instagram influencers pay us money to take pictures with our flowers
- We want to maximize our blooms, so we decide to run an experiment to determine which soil and fertilizer type increases our bloom density

Texas Bluebonnets



- 2 soil types (green fill), 3 fertilizer types (outline)
- every combination 5 times
- randomly distribute treatment plots throughout our field



Switch to ze code!